

PARTS LIST

Semiconductors:

- IC1 - 74HC14 hex inverter
- LED1-LED5 - 5mm red LED
- D1-D4 - 1N4001 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 2.2-kilo-ohm
- R2 - 1.8-kilo-ohm
- R3, R5, R6, R8, R9 - 1-kilo-ohm
- R4 - 680-ohm
- R7 - 470-ohm

Capacitors:

- C1-C5 - 100 μ F, 25V electrolytic
- C6 - 0.33 μ F ceramic disk

Miscellaneous:

- CON1 - 6-pin connector
- MCU

N4, resistor R3 and capacitor C3. The direction of diode D3 as indicated in the previous arrangements.

If you want two LEDs blinking alternately, the setup is made using inverters N5 and N6 together with resistor R4 and capacitor C4.

When OUT3 is low, LED4 is 'off' and LED5 is 'on.' When OUT3 is high, LED4 and LED5 blink alternately. The frequency of blinking in all the cases

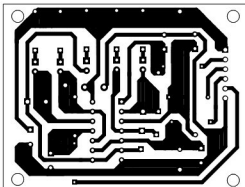


Fig. 2: An actual-size, single-side PCB for efficient LED blinking for embedded systems

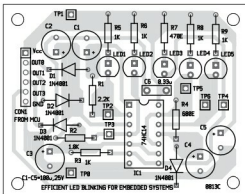


Fig. 3: Component layout for the PCB

depends on the value of the adjoining resistor and capacitor.

Construction and testing

An actual-size, single-side PCB for efficient LED blinking for embedded systems is shown in Fig. 2 and its component layout in Fig. 3. The PCB provided is only for learning purpose and can be enclosed in a suitable case.

To test the circuit, measure input supply at TP1. It should be around 5V DC with respect to TP0. Now make OUT0 high and measure the frequency at TP2. It should be around 2.5 Hz. Similarly, measure the frequencies at test points TP3, TP4, TP5 and TP6 as shown in the test points table. ●

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